Inorganic Cumulative Exam

March 11, 2011

Several "bonding theories" have been developed to help inorganic chemistry deal with the seemingly overwhelming diversity of inorganic compounds. Three of these that show a very high level of success are addressed in the following questions

Valence Shell Electron-Pair Repulsion (VESPR). Give an example of a molecule with a central atom surrounded by 5 peripheral atoms or groups. Describe the geometry including specific angles. Are there isomers in the case of molecules with the general formulation MX_3Y_2? If so how what "rules" are used for these? How would you describe the success of this "theory" in predicting structures

Wades Rules. These rules are based on three basic categories. Describe these categories. Give specific molecular examples of applications of this set of rules to molecules that fall into each of the basic categories. What does the term isolobal mean?

18-Electron Rule. Give two examples, one compound that fits the rules and one that is not expected to obey the rule, based on the general basis of the rules. What is the significance of the number 18 in terms of molecular orbitals.